



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Monday, July 6, 2026



URL of the GitHub repository including all the
completed notebooks and Python files

<https://github.com/mzorrellacampos/IBM-Data-Science-Capstone>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Objective

- Develop a machine learning model capable of predicting whether the first stage of a Falcon 9 rocket will successfully land after launch.

Summary of Methodology

- Collected launch data from the SpaceX REST API and complementary information through web scraping.
- Cleaned and prepared the dataset using data wrangling techniques.
- Performed exploratory data analysis using visualizations and SQL.
- Built interactive visualizations with Folium and Plotly Dash.
- Developed and compared multiple classification models to predict landing success.

Summary of Key Results

- Launch success is strongly influenced by launch site, orbit type, payload mass, and flight history.
- Interactive dashboards helped identify important operational patterns across launch sites.
- Among the evaluated models, the best classifier achieved the highest predictive accuracy for landing outcomes.

Introduction

Project Background and Context

- SpaceX significantly reduces launch costs by recovering and reusing the Falcon 9 first-stage booster. Being able to accurately predict whether the booster will successfully land is valuable for mission planning, operational decision-making, and cost estimation.

Business Problem to Find Answers To

- This project aims to identify the factors associated with successful first-stage landings and build a predictive model capable of estimating landing outcomes using historical launch data.

Main Questions

- Which mission characteristics most influence landing success?
- How do launch site, orbit type, payload mass, and flight number affect the probability of a successful landing?
- Which classification algorithm provides the best predictive performance?

Section 1

Methodology

Methodology

Data Collection Methodology

- Retrieve launch records from the SpaceX REST API.
- Collect complementary information through web scraping.

Data Wrangling

- Clean, merge, and transform the collected datasets.
- Handle missing values and prepare features for analysis.

Exploratory Data Analysis (EDA) Using Visualization and SQL

- Explore relationships between launch outcomes and mission variables.
- Perform SQL analysis and statistical visualizations.

Interactive Visual Analytics Using Folium and Plotly Dash

- Build interactive maps using Folium.
- Develop a Plotly Dash dashboard for dynamic data exploration.

Predictive Analysis Using Classification Models

- Train multiple classification models.
- Tune hyperparameters and compare model performance.
- Select the best-performing classifier.

Data Collection

Objective

- Collect historical Falcon 9 launch data required for exploratory and predictive analyses.

Data Sources

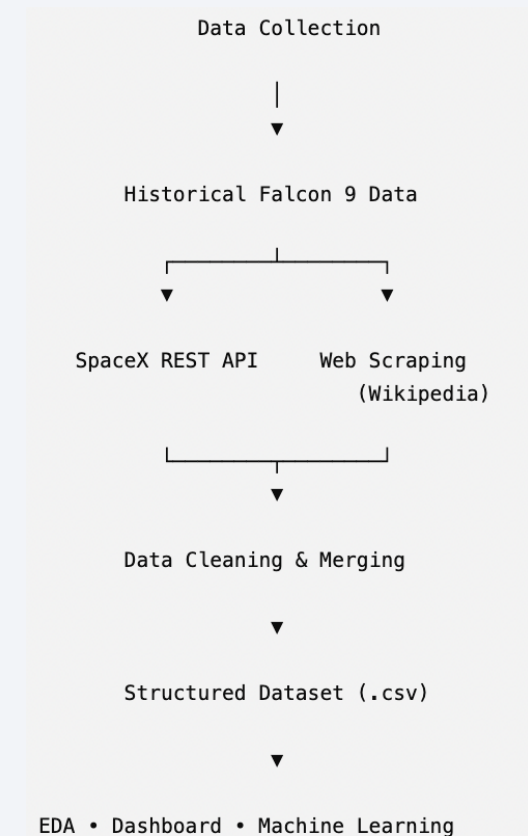
- SpaceX REST API
- Public SpaceX web pages

Presentation of the Collected Data

- Launch dates
- Launch sites
- Booster versions
- Orbit types
- Payload mass
- Landing outcomes
- Mission success

Output

- A structured dataset containing historical launch records ready for preprocessing.



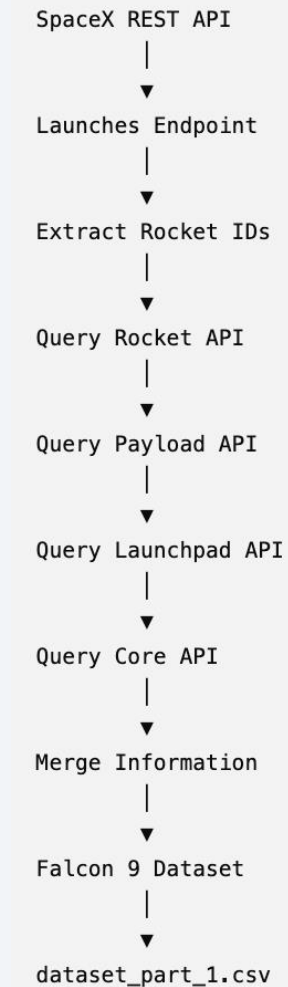
Data Collection – SpaceX API

SpaceX REST API

- Retrieved mission records directly from the official SpaceX REST API.
- Queried multiple API endpoints to obtain launch, rocket, payload, core, and launch pad information.
- Combined the responses into a unified launch dataset.
- Exported the processed dataset for downstream analysis.

GitHub URL Reference

<https://github.com/mzorrellacampos/IBM-Data-Science-Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb>



Data Collection - Scraping

Web Scraping

- Collected complementary launch information unavailable through the API.
- Extracted tabular data from publicly available web pages.
- Parsed HTML tables using BeautifulSoup and Pandas.
- Combined scraped data with the API dataset to improve completeness.

GitHub URL Reference

<https://github.com/mzorillacampos/IBM-Data-Science-Capstone/blob/main/jupyter-labs-webscraping.ipynb>



Data Wrangling

Data Preparation Process

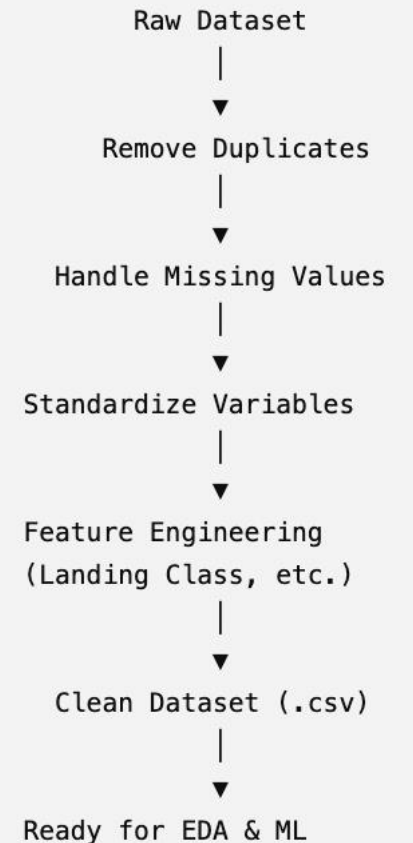
- Removed duplicated records.
- Managed missing values.
- Standardized categorical variables.
- Selected relevant features for analysis.
- Created binary landing outcome labels.
- Prepared the final dataset for visualization and machine learning.

Output

- A clean and analysis-ready dataset suitable for EDA and predictive modeling.

GitHub URL Reference

<https://github.com/mzorillacampos/IBM-Data-Science-Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb>



EDA with Data Visualization

Visualizations Performed

- Flight Number vs Launch Site
- Payload Mass vs Launch Site
- Success Rate by Orbit Type
- Flight Number vs Orbit Type
- Payload Mass vs Orbit Type
- Yearly Launch Success Trend

Purpose

- These visualizations were used to identify relationships between mission characteristics and landing success, detect trends, and support feature selection for predictive modeling.

GitHub URL Reference

<https://github.com/mzorillacampos/IBM-Data-Science-Capstone/blob/main/jupyter-labs-eda-dataviz.ipynb>

EDA with SQL

SQL Analysis

- SQL queries were used to explore operational characteristics of SpaceX launches.
- The analyses included:
- Launch site identification
- Payload aggregation
- Average payload by booster version
- Landing success history
- Booster performance
- Mission outcome statistics
- Landing outcome ranking
- Historical launch records
- These queries complemented the visual analysis by providing aggregated operational insights directly from the relational dataset.

GitHub URL Reference

https://github.com/mzorrellacampos/IBM-Data-Science-Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

Interactive Geographic Analysis

- The Folium application was developed to visualize launch locations and their surrounding infrastructure.

Map Elements

- Launch site markers
- Color-coded landing outcomes
- Coastlines
- Railways
- Highways
- Nearby cities
- Distance measurements

Objective

- To evaluate whether geographical characteristics surrounding launch sites could help explain landing performance.

GitHub URL Reference

https://github.com/mzorillacampos/IBM-Data-Science-Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Interactive Dashboard

- The dashboard allows dynamic exploration of launch data.

Dashboard Components

- Launch success pie charts
- Payload range selector
- Scatter plots
- Launch site filters

Purpose

- Enable users to interactively explore how payload mass, launch site, and booster characteristics affect landing success.

GitHub URL Reference

<https://github.com/mzorillacampos/IBM-Data-Science-Capstone/blob/main/spacex-dash-app.py>

Predictive Analysis (Classification)

Classification Models Evaluated

- Logistic Regression
- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors

Evaluation Metrics

- Accuracy
- Confusion Matrix
- Cross-validation performance
- Best model selection

GitHub URL Reference

https://github.com/mzorrellacampos/IBM-Data-Science-Capstone/blob/main/NOMBRE_EXACTO_DEL_ARCHIVO.ipynb

Model Development Workflow

- Dataset
- ↓
- Train/Test Split
- ↓
- Feature Scaling
- ↓
- Model Training
- ↓
- Hyperparameter Tuning
- ↓
- Model Evaluation
- ↓
- Best Model Selection

Results

Exploratory Data Analysis

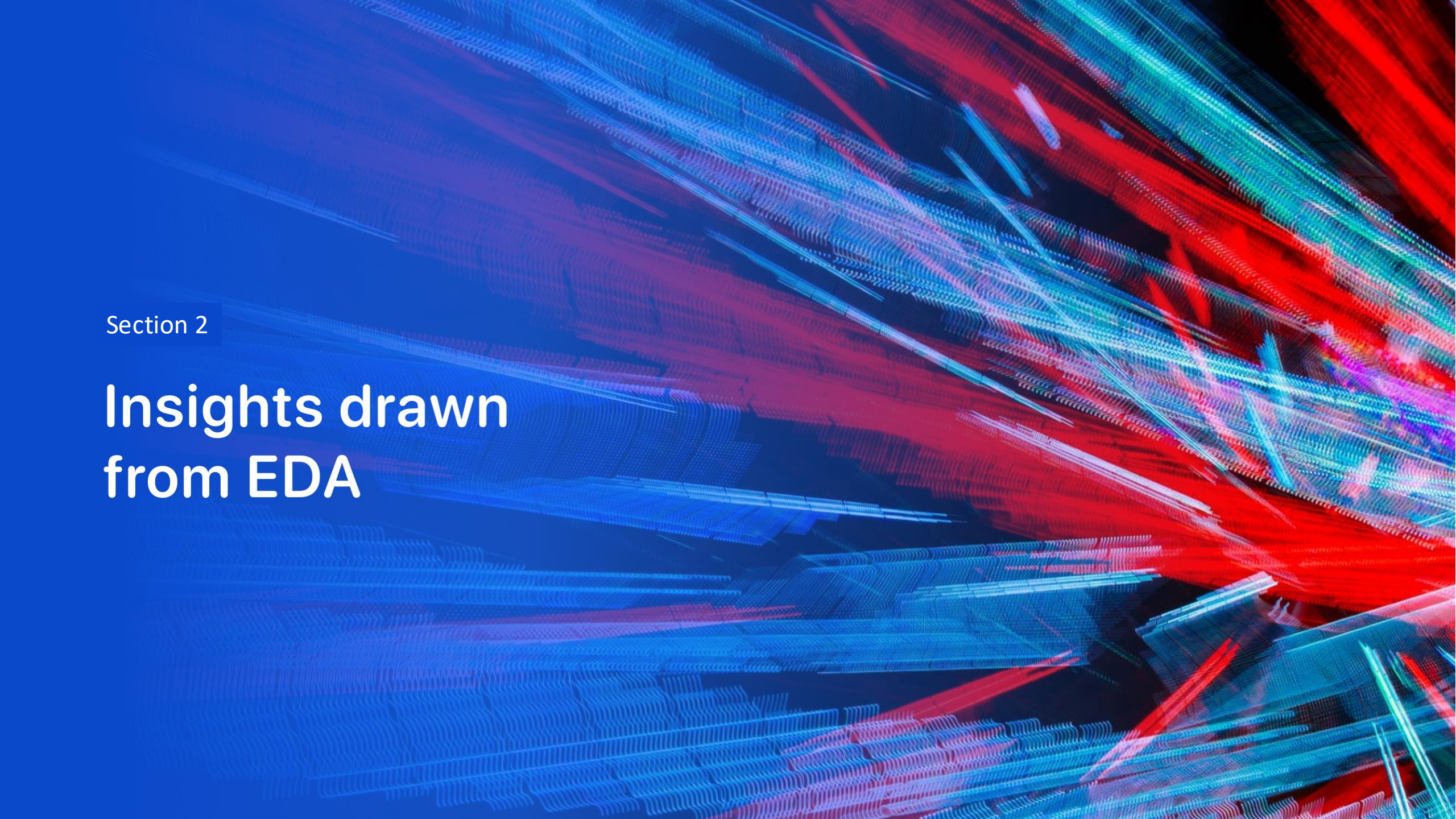
- Identified clear relationships between launch success, payload mass, orbit type, and launch site.
- Launch success improved consistently over time as SpaceX gained operational experience.
- Certain orbit types and launch sites achieved higher success rates than others.

Interactive Visual Analytics

- Interactive Folium maps were created to visualize launch sites, landing outcomes, and geographical proximity to infrastructure.
- Plotly Dash dashboards enabled dynamic exploration of launch performance across launch sites and payload ranges.

Predictive Analysis

- Multiple classification models were trained and evaluated, including Logistic Regression, SVM, Decision Tree, and KNN.
- Logistic Regression achieved the best overall performance, with approximately **84.6% cross-validation accuracy** and **83.3% test accuracy**.
- The final model demonstrated reliable predictive capability for first-stage landing success.

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

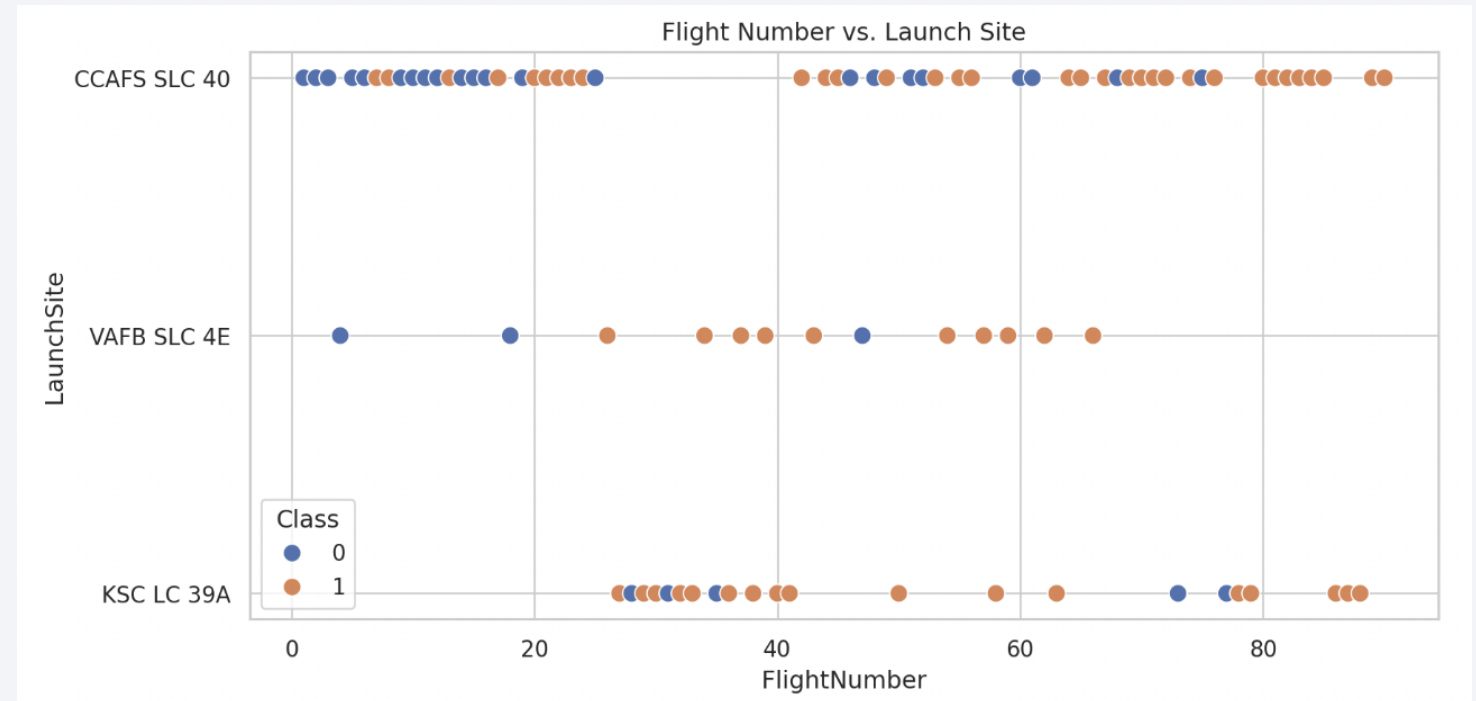
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Explanation:

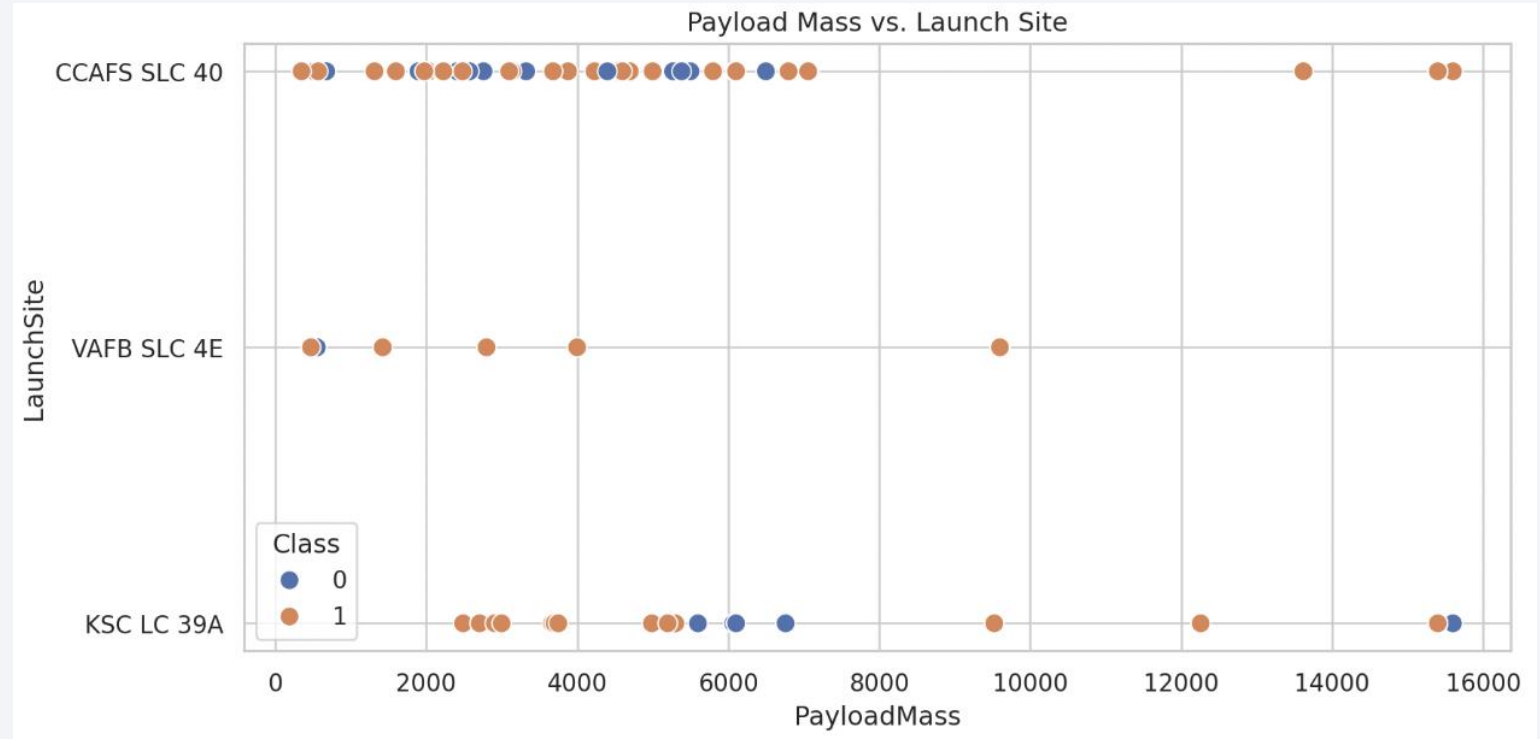
- This scatter plot shows the relationship between flight number and launch site.
- Launches became more frequent over time, particularly from **CCAFS LC-40** and **KSC LC-39A**.
- The distribution illustrates how SpaceX expanded its launch operations while maintaining activity across multiple launch facilities.



Payload vs. Launch Site

Explanation:

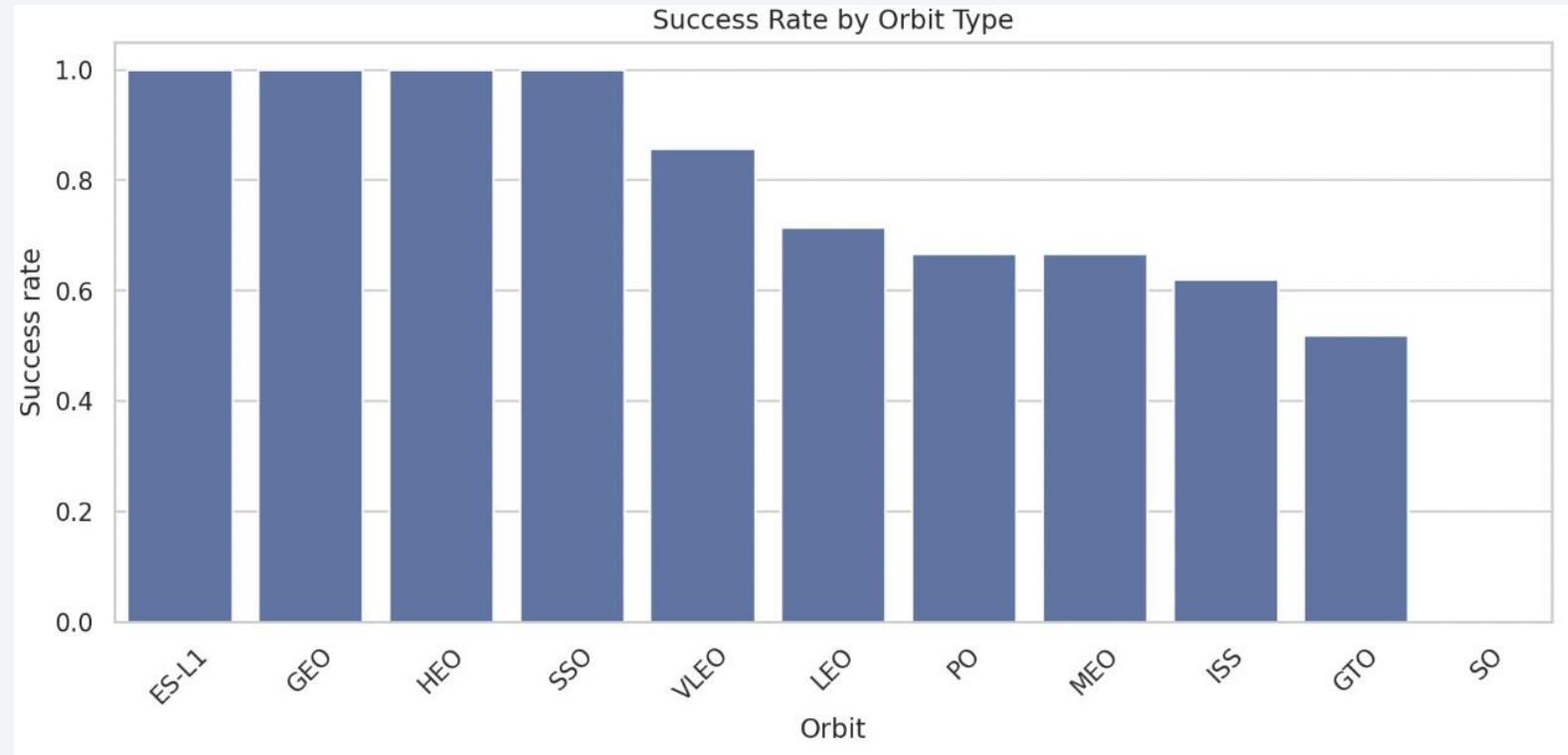
- The scatter plot compares payload mass across different launch sites.
- Larger payloads were generally launched from **KSC LC-39A** and **CCAFS LC-40**.
- The visualization suggests that launch site selection is related to mission characteristics and payload requirements.



Success Rate vs. Orbit Type

Explanation:

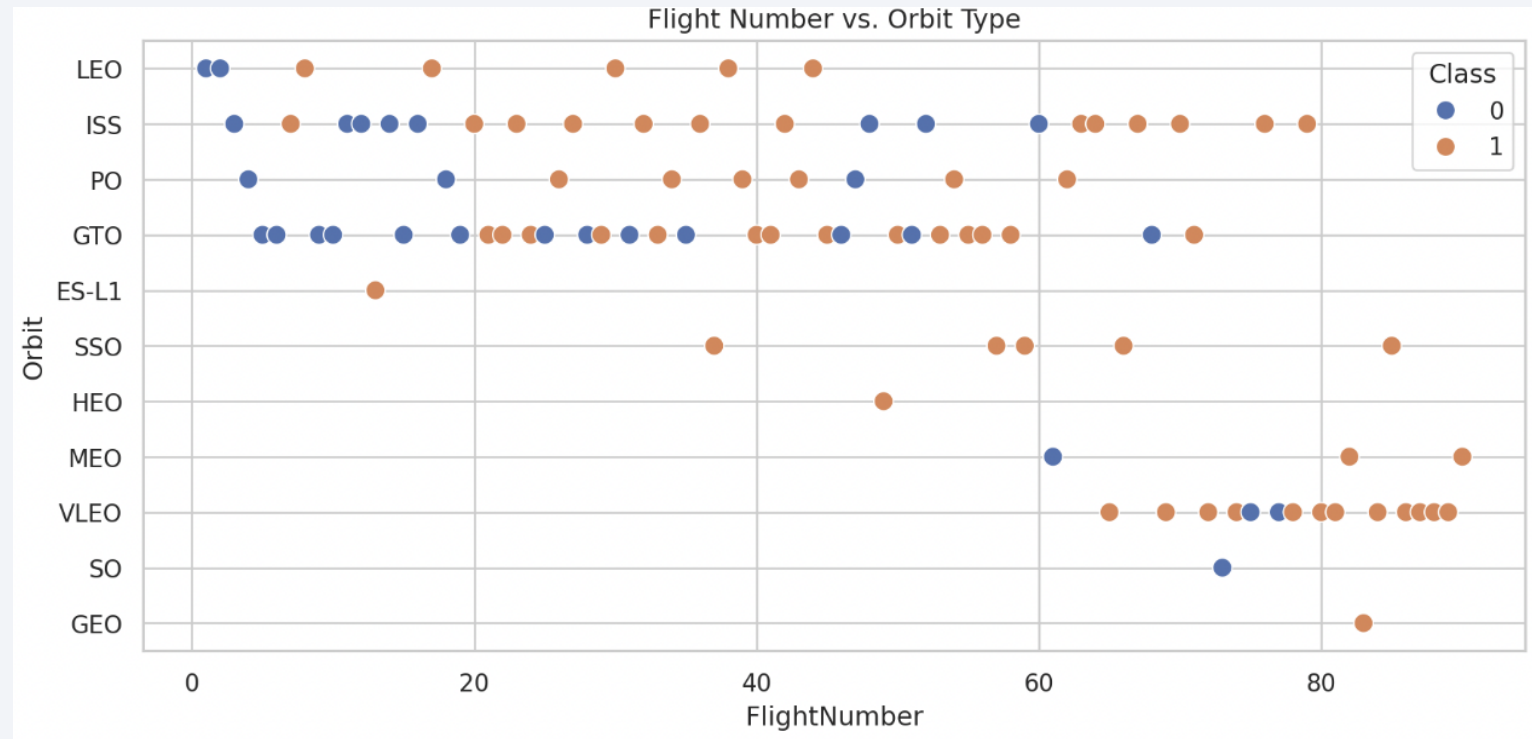
- This bar chart compares launch success rates among orbit types.
- Some orbit categories achieved consistently higher success rates than others.
- The results indicate that mission profile and orbit destination influence landing success probability.



Flight Number vs. Orbit Type

Explanation:

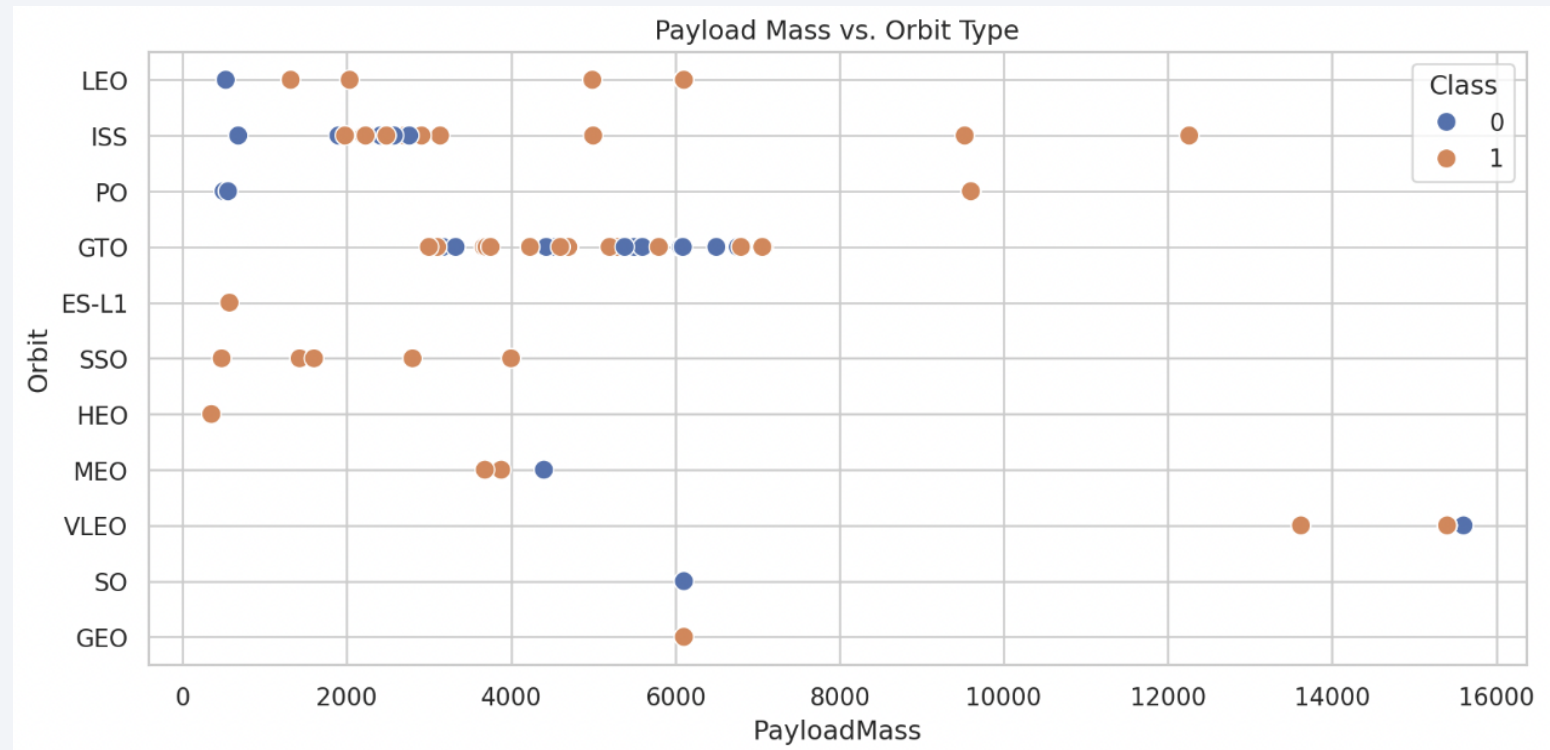
- This scatter plot relates flight number to orbit type.
- As flight numbers increased, SpaceX successfully completed missions to a wider variety of orbit types.
- The trend reflects the company's growing operational experience and mission diversity.



Payload vs. Orbit Type

Explanation:

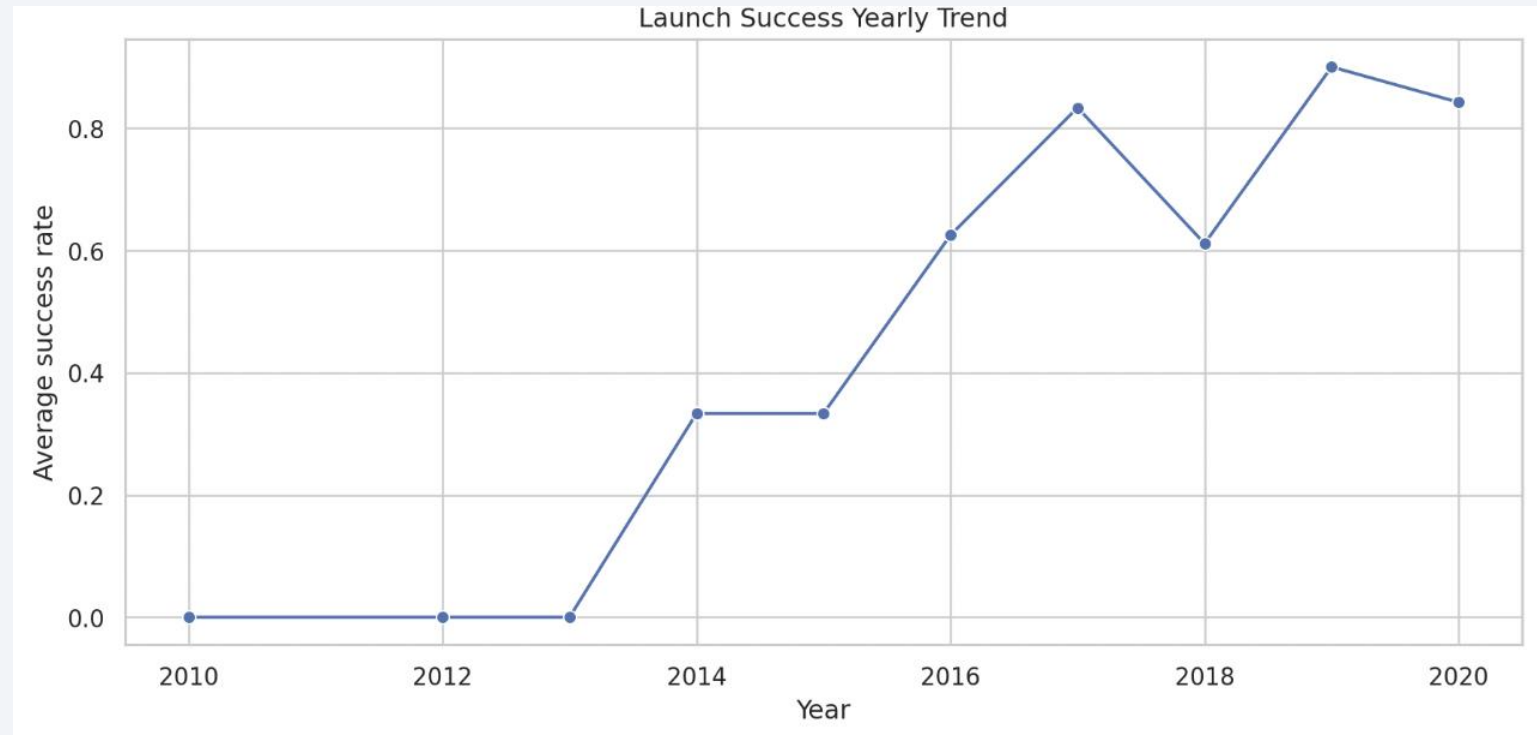
- This chart compares payload mass with orbit type.
- Heavier payloads are generally associated with specific mission profiles such as **GTO**.
- The visualization highlights the relationship between payload requirements and orbital destination.



Launch Success Yearly Trend

Explanation:

- The yearly success rate shows a clear upward trend over time.
- Early missions experienced lower success rates, while recent years achieved consistently higher performance.
- This reflects continuous improvements in launch technology, operational procedures, and booster recovery capabilities.



All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Explanation

- This query retrieves all unique launch site names used in the SpaceX dataset.
- The results confirm that SpaceX operated from four primary launch facilities during the study period.
- Identifying launch sites provides the basis for comparing launch frequency and success rates across locations.

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Show 10 per page

Explanation

- This query filters launch sites whose names begin with "CCA".
- The results identify launches performed from Cape Canaveral facilities.
- Filtering data using SQL conditions enables focused analysis of specific launch locations.

Total Payload Mass

total_payload_mass_kg
45596
Show 10  per page

Explanation

- This query calculates the total payload mass delivered during NASA CRS missions.
- The result summarizes the total cargo transported for NASA resupply operations.
- Aggregate SQL functions provide an efficient way to summarize mission performance.

Average Payload Mass by F9 v1.1

avg_payload_mass_kg
2928.4
Show 10  per page

Explanation

- This query computes the average payload mass carried by the Falcon 9 v1.1 booster.
- The average provides an indication of the typical mission capacity for this booster version.
- Comparing averages across booster versions helps evaluate vehicle evolution.

First Successful Ground Landing Date

first_success_ground_landing_date
2015-12-22
Show 10  per page

Explanation

- This query identifies the first successful landing on a ground pad.
- The milestone represents a major achievement in reusable rocket technology.
- SQL date functions make it easy to identify significant historical events.

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2
Show 10 per page

Explanation

- This query lists missions that successfully landed on a drone ship while carrying payloads between 4,000 and 6,000 kg.
- The results demonstrate SpaceX's ability to recover boosters during medium-to-heavy payload missions.
- Combining multiple filtering conditions allows detailed mission selection.

Total Number of Successful and Failure Mission Outcomes

Mission_Outcome	count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Show 10 per page

Explanation

- This query summarizes successful and unsuccessful landing outcomes.
- Successful landings occur more frequently than failures in the analyzed dataset.
- The results illustrate the significant improvement achieved in reusable launch operations.

Boosters Carried Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6

Show 10 per page

Explanation

- This query identifies the booster(s) that carried the maximum payload.
- These missions represent the highest payload capability achieved during the analyzed period.
- Identifying extreme values is useful for benchmarking operational performance.

2015 Launch Records

month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Show per page

Explanation

- This query lists unsuccessful drone ship landing attempts during 2015.
- The results provide insight into the challenges faced while developing booster recovery technology.
- Reviewing historical failures helps understand subsequent improvements.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Landing_Outcome	count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Show 10 per page

Explanation

- This query ranks landing outcomes by frequency over the selected period.
- "No attempt" and successful recoveries appear among the most common outcomes.
- Ranking frequencies provides an overall picture of landing performance trends.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

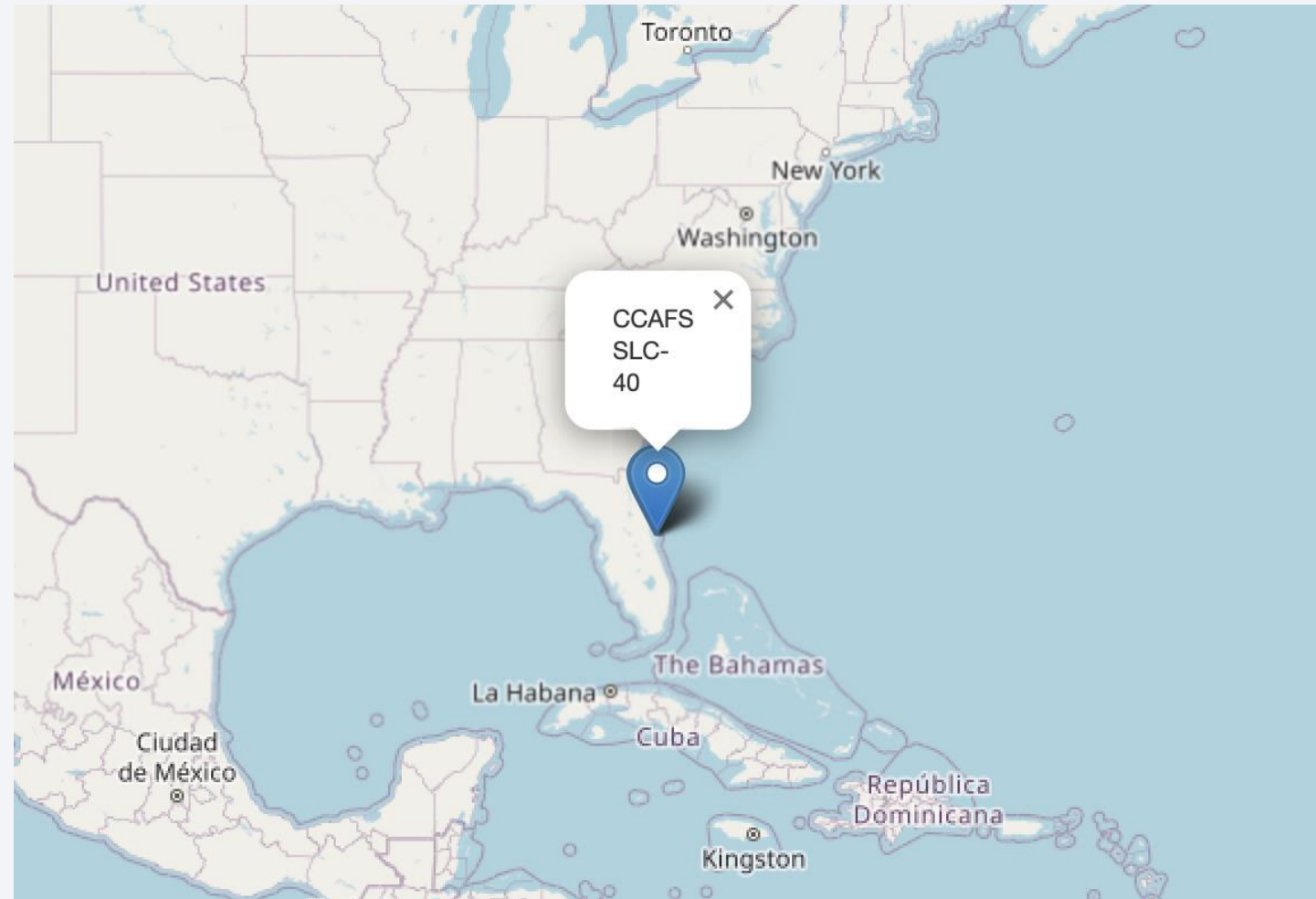
Global Distribution of SpaceX Launch Sites

Explanation

- This interactive map displays the geographical locations of all SpaceX launch sites included in the analysis.
- Each marker represents one launch facility used during the study period.
- The map provides geographical context for comparing launch operations across different locations.

Key Findings

- Most launches were concentrated on the east coast of the United States.
- Launch sites are strategically located near the Atlantic Ocean, providing safe launch corridors.
- The distribution reflects SpaceX's operational expansion over time.



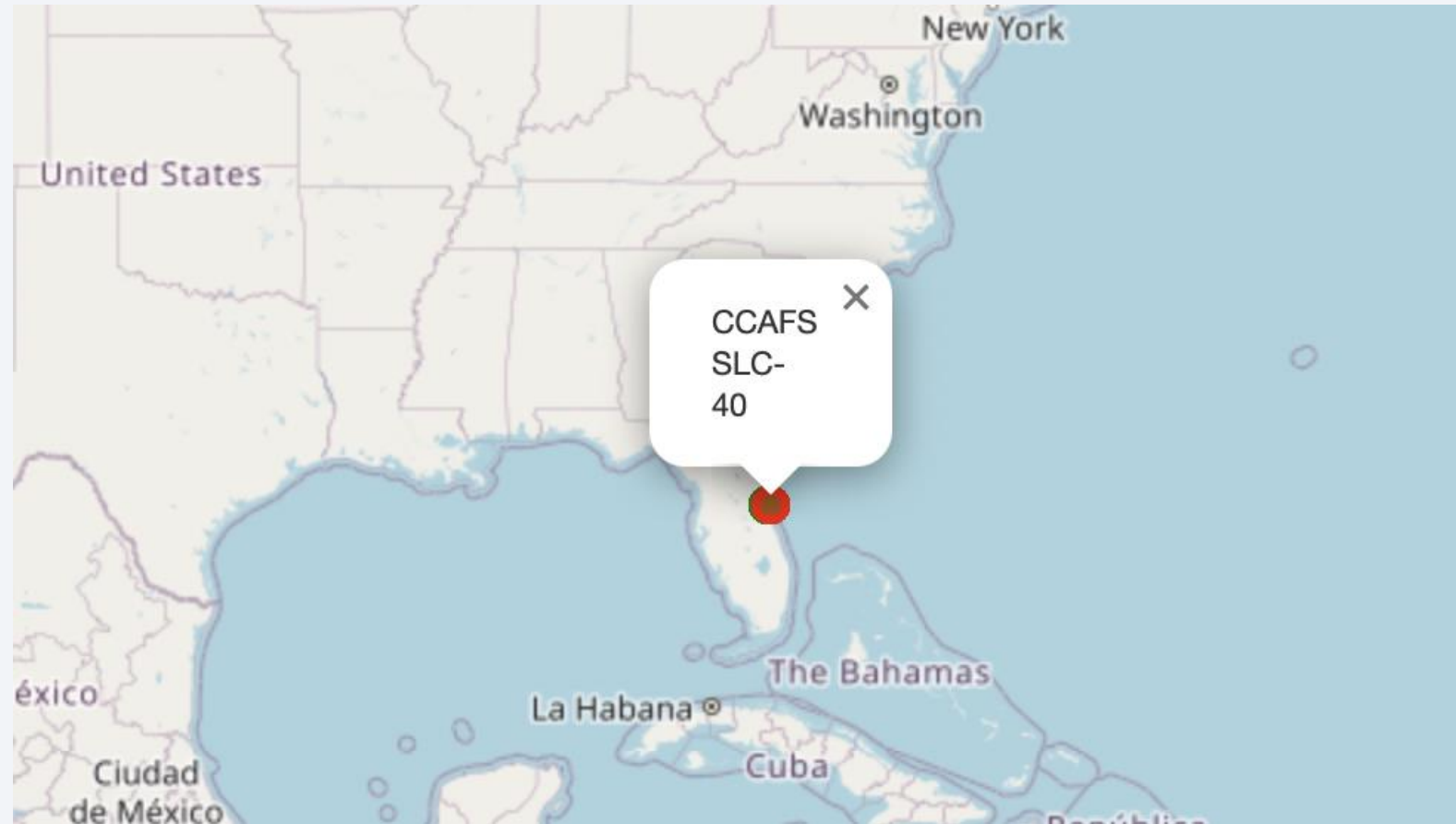
Launch Outcomes by Launch Site

Explanation

- Marker colors distinguish successful and unsuccessful landing outcomes.
- The interactive visualization makes it easy to identify patterns in landing performance across launch sites.
- Spatial visualization complements the statistical analysis performed during EDA.

Key Findings

- Successful landings become increasingly frequent in later missions.
- Multiple launch sites exhibit consistently high landing success rates.
- Geographic visualization confirms trends observed in the statistical analyses.



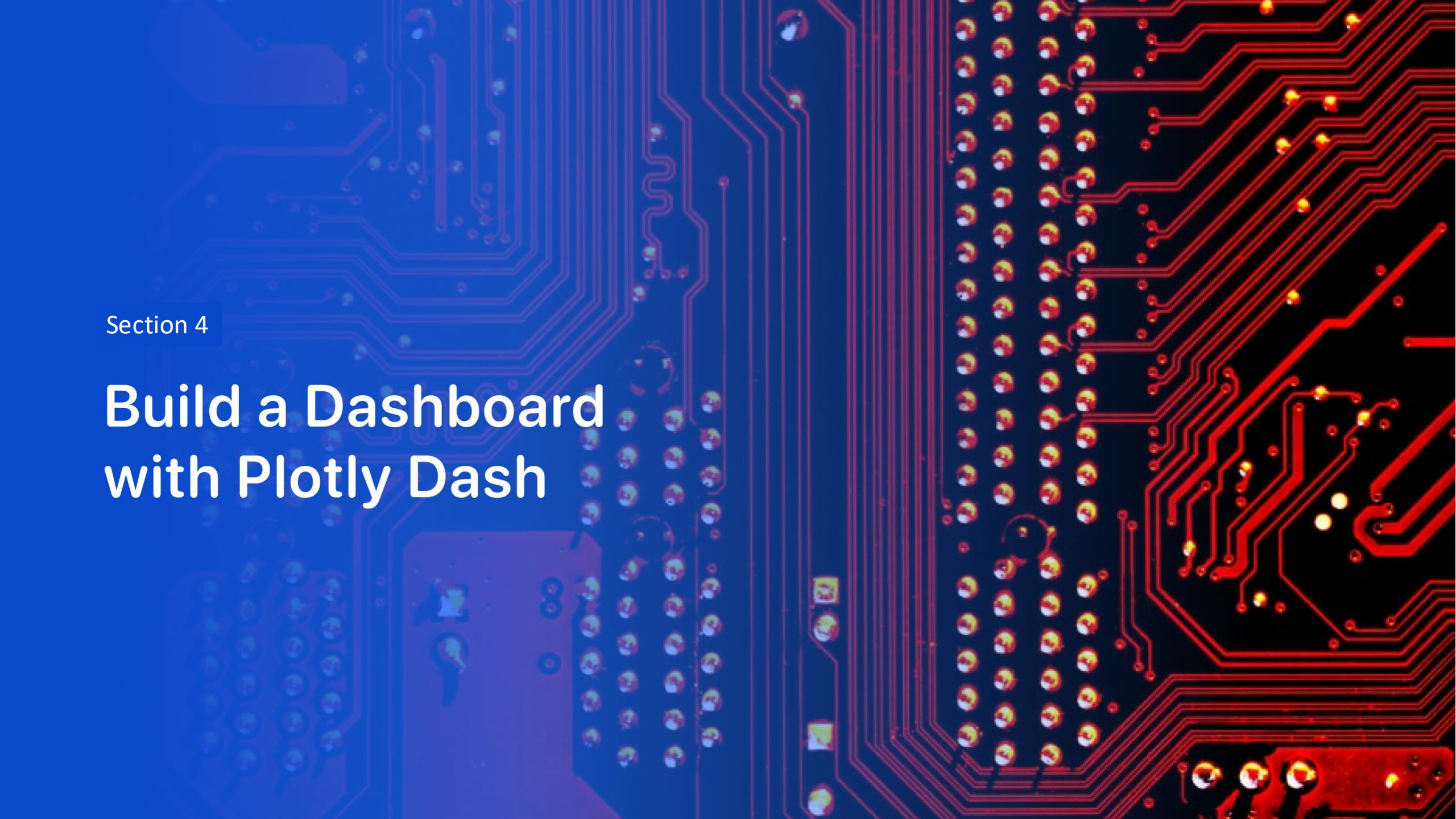
Interactive Dashboard

The Plotly Dash dashboard was designed to provide an interactive environment for exploring SpaceX launch data.

Key interactive features include:

- Filtering launches by launch site.
- Selecting payload mass ranges using a slider.
- Viewing launch success distribution through pie charts.
- Exploring the relationship between payload mass and launch success using scatter plots.
- Supporting dynamic exploration of launch performance across different missions.





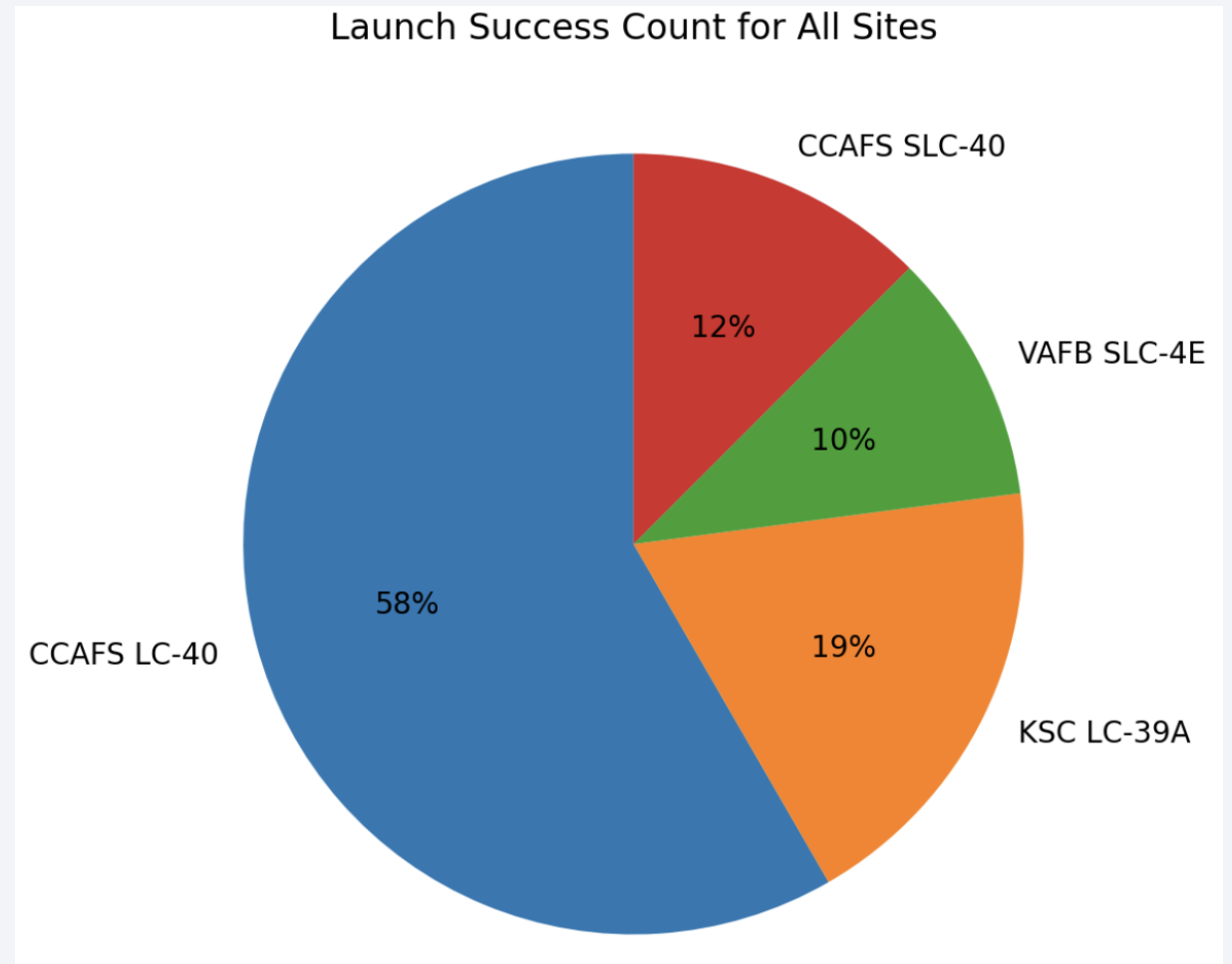
Section 4

Build a Dashboard with Plotly Dash

Launch Success for All Sites

Explanation

- This pie chart summarizes the distribution of successful launches across all launch sites.
- CCAFS LC-40 accounts for the largest share of successful missions.
- The visualization provides an overall comparison of launch activity among SpaceX launch sites.



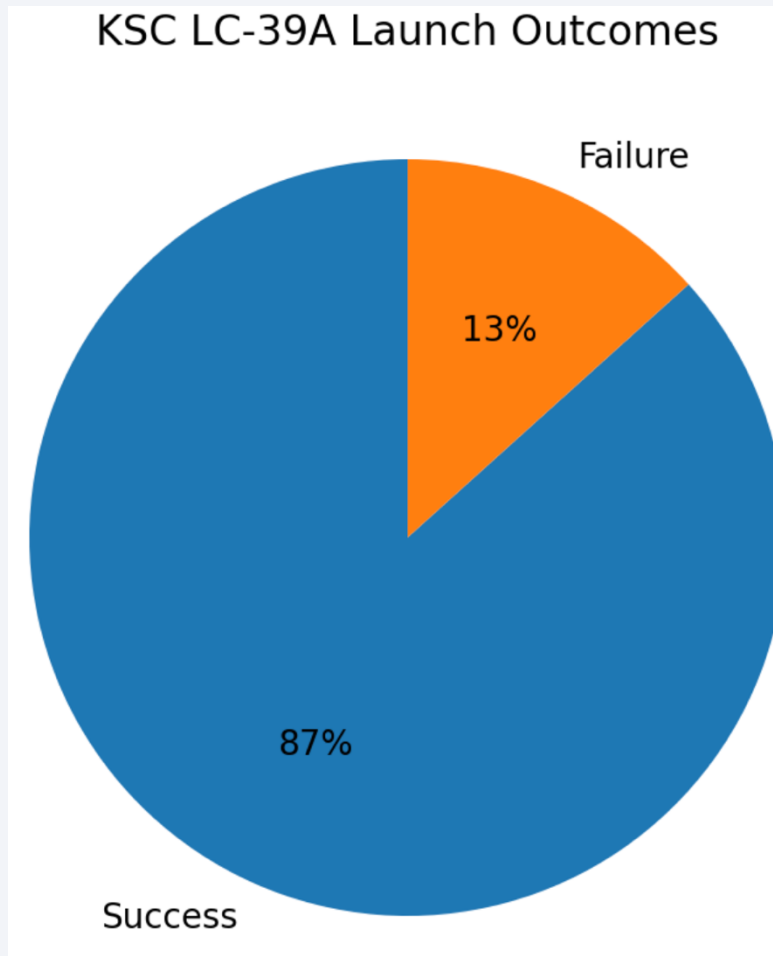
Launch Success Ratio at the Best Performing Site

Explanation

This pie chart focuses on the launch site with the highest success ratio.

Most launches from this site resulted in successful first-stage landings.

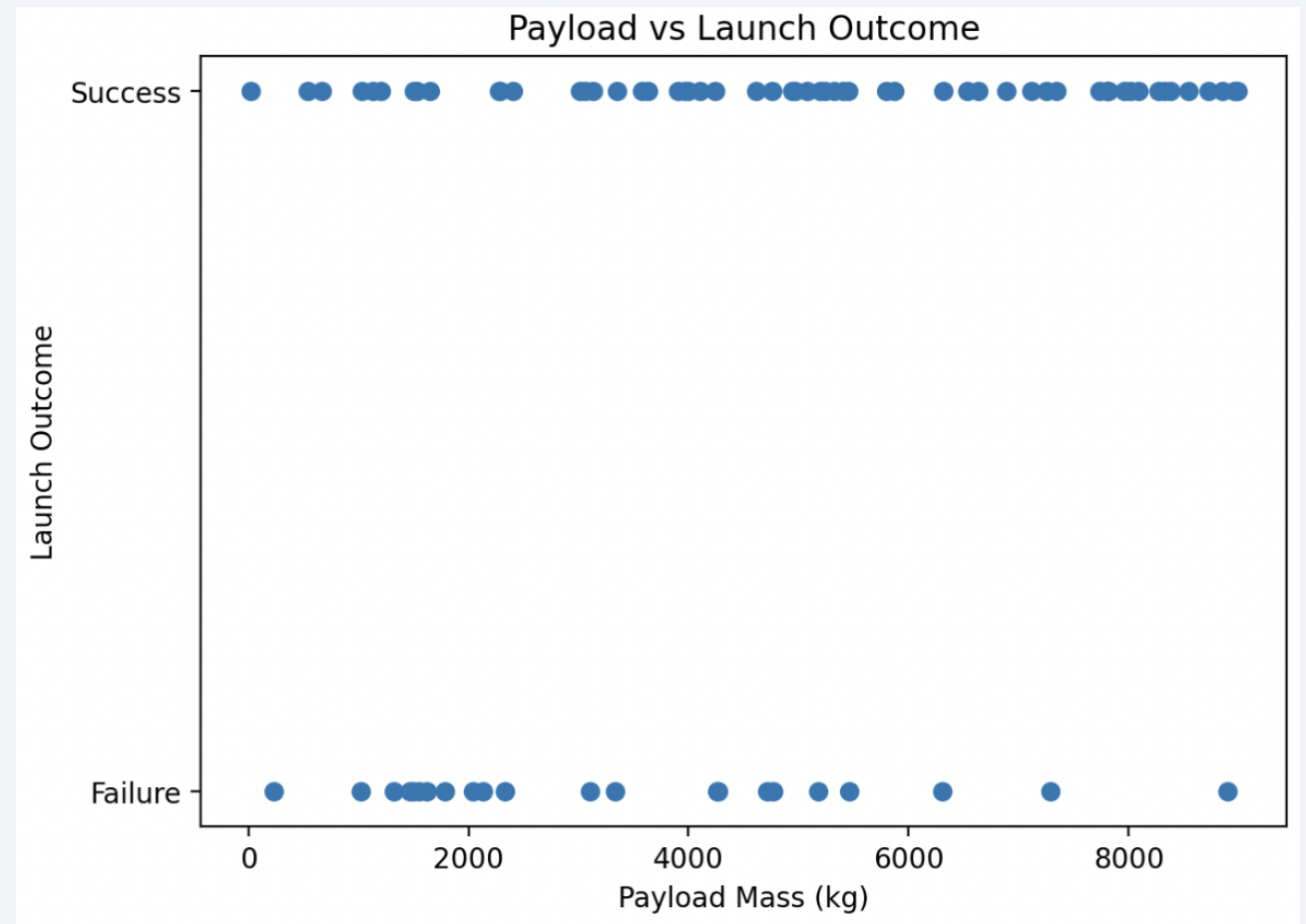
The visualization highlights the operational reliability achieved at this launch facility.



Payload vs. Launch Outcome

Explanation

- This scatter plot illustrates the relationship between payload mass and launch outcome.
- Successful landings occur across a wide range of payload masses.
- Interactive filtering by payload range helps identify mission profiles associated with higher landing success.





Section 5

Predictive Analysis (Classification)

Classification Accuracy

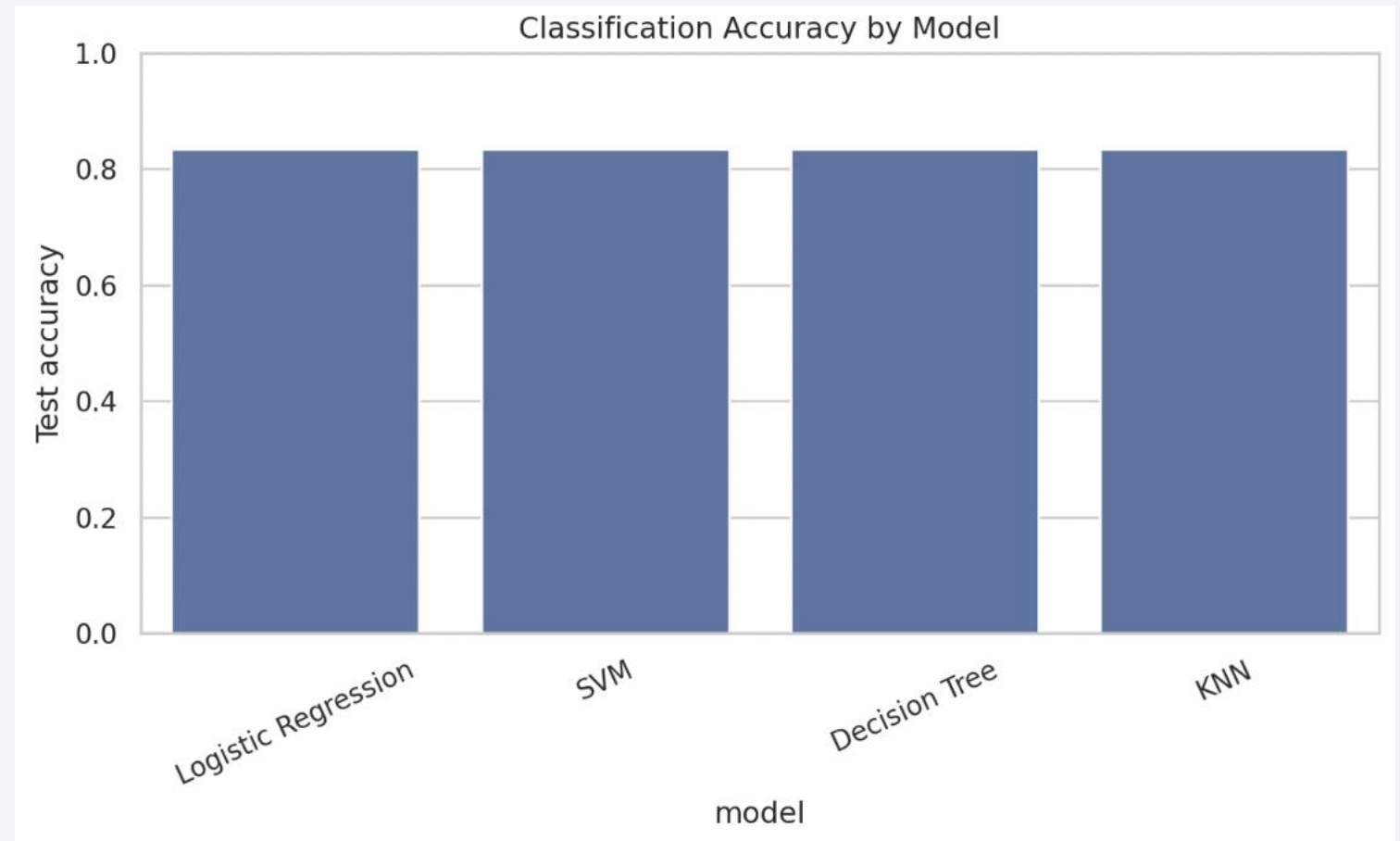
Explanation

- Four classification algorithms were evaluated:
 - Logistic Regression
 - Support Vector Machine
 - Decision Tree
 - K-Nearest Neighbors
- Models were compared using classification accuracy.
- Logistic Regression achieved the highest overall predictive performance.

Key Result

Best model: Logistic Regression

Test Accuracy: 83.33%



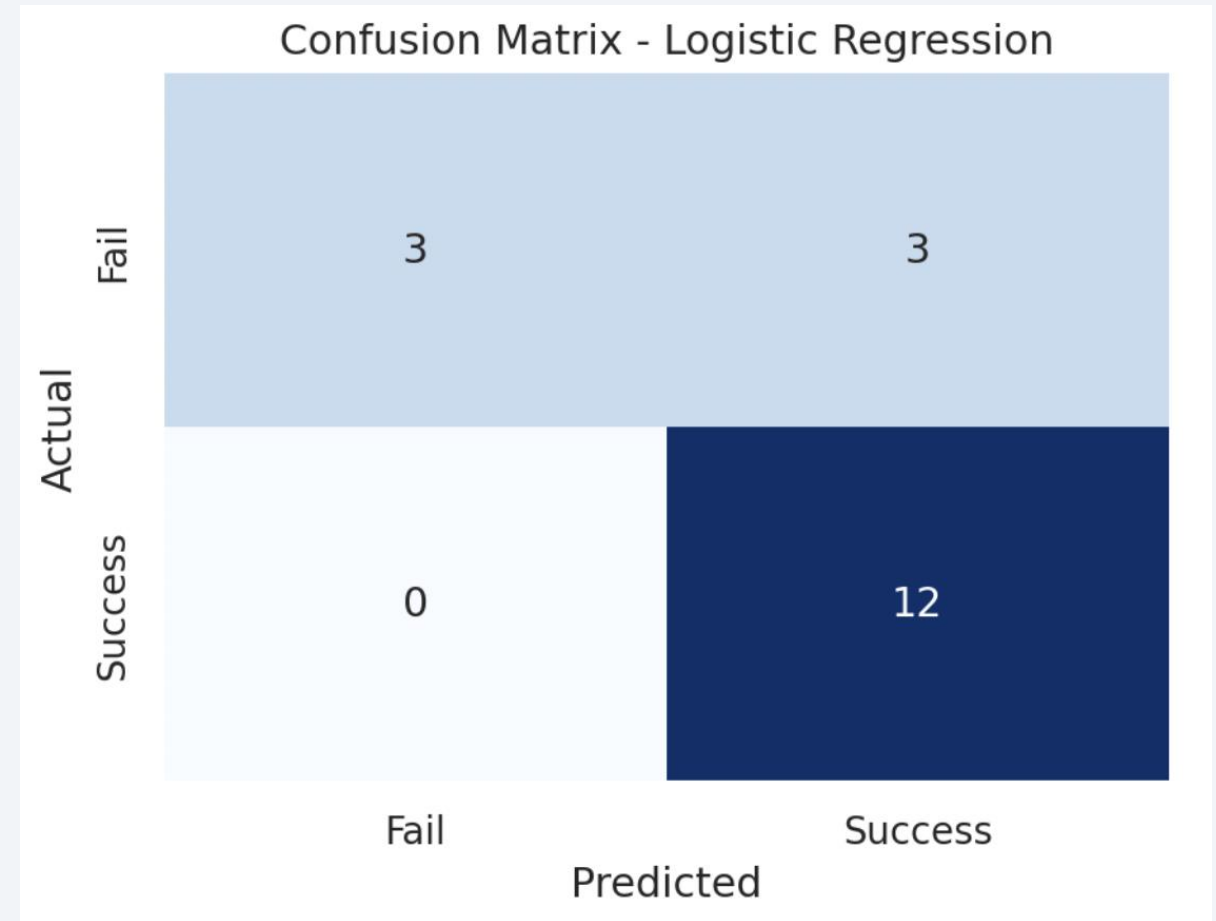
Confusion Matrix

Explanation

- The confusion matrix summarizes the model's prediction performance.
- Most landing outcomes were classified correctly.
- Misclassifications were limited, indicating good generalization on unseen data.

Interpretation

- High number of True Positives and True Negatives.
- Few False Positives and False Negatives.
- The model demonstrates good predictive reliability for first-stage landing success.



Conclusions

- Historical launch data reveal strong relationships between landing success, launch site, orbit type, payload mass, and flight number.
- Exploratory analysis and SQL queries provided valuable insights into SpaceX operational performance.
- Interactive visualizations facilitated exploration of geographical and mission-related patterns.
- Among the evaluated machine learning algorithms, **Logistic Regression** achieved the best predictive performance (**83.33% test accuracy**).
- The developed workflow demonstrates how data science techniques can effectively support prediction of Falcon 9 first-stage landing success.

Appendix

- Python notebooks used for data collection, exploratory data analysis, visualization, and machine learning.
- SQL queries for launch statistics, payload analysis, and mission performance.
- Interactive Folium maps showing launch sites and landing outcomes.
- Plotly visualizations and dashboard components for interactive exploration.
- Machine learning evaluation results, including confusion matrix and model comparison.
- Project datasets and generated CSV output files.

Thank you!

